

Validation of the features on the Goodscents dataset

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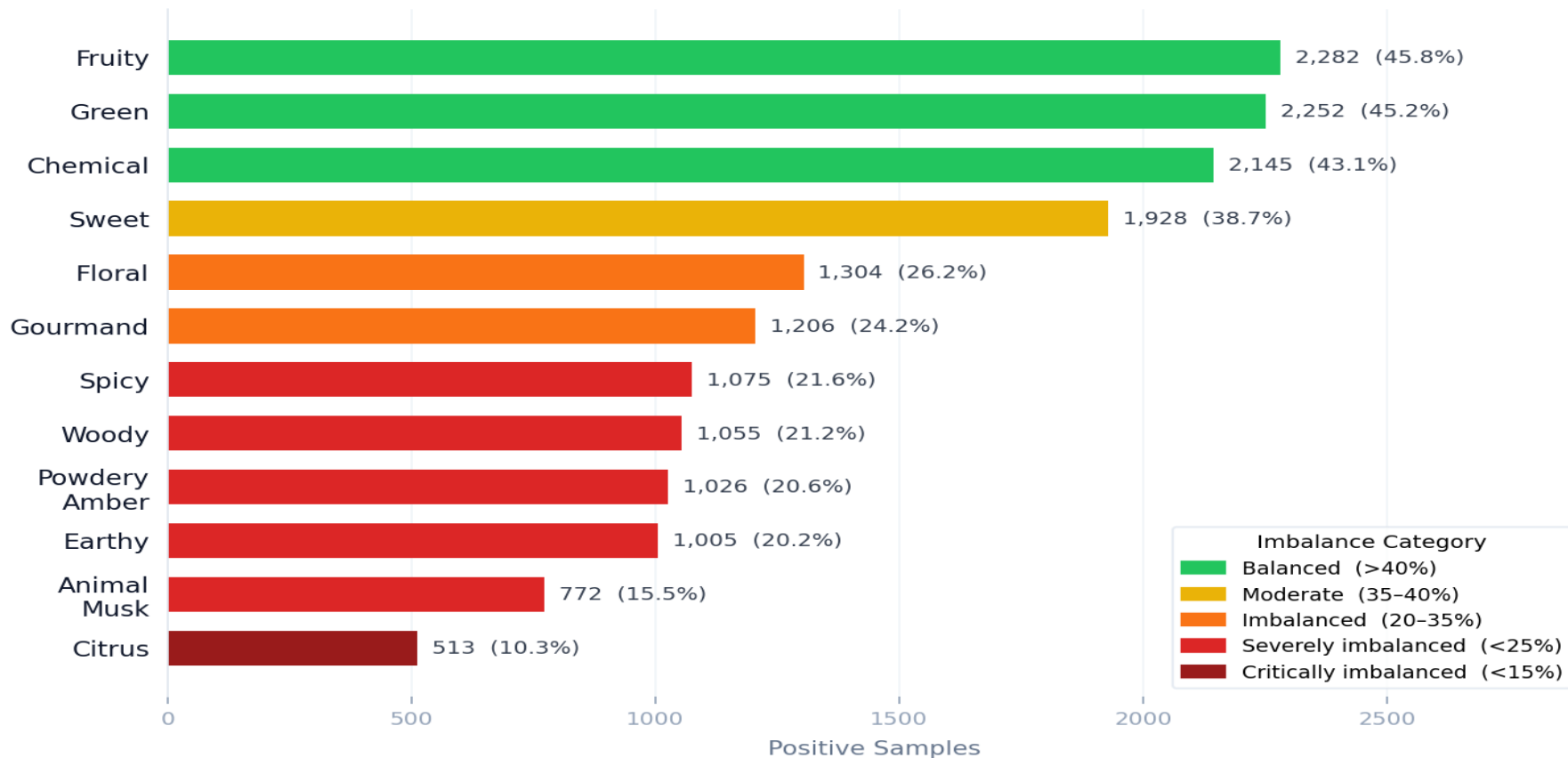
2.1 Dataset Overview



Why Reduce to 12 Labels?



2.2 Label Distribution & Imbalance



The Imbalance Problem : Animal Musk Example

Animal Musk:

772 positive out of 4,981 total

only 15.5% of the dataset

Severely imbalanced

87.9%

Accuracy

Looks good...

35.4%

True Positive Rate

65% of musks are missed!

66.5%

Balanced Accuracy

Reveals the real picture

47.2%

F1 Score

Poor minority performance

Why this matters beyond bad metrics

The model learns to predict **not Animal Musk** most of the time which is correct 84.5% of the time by default.

Feature selection becomes corrupted: selected features characterize the majority class, not Animal Musk chemistry.

This is not just a performance problem , it is a fundamental validity problem.

SMOTE : Synthetic Minority Over-sampling Technique



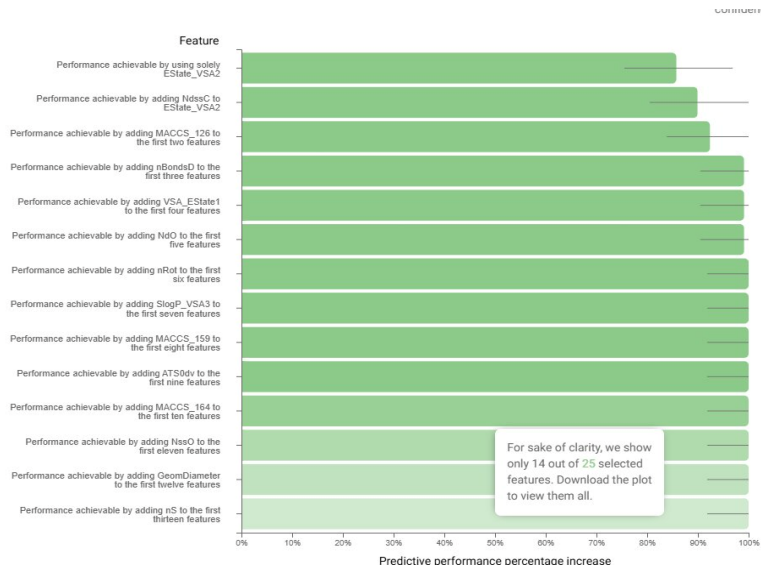
$$x_{\text{new}} = x_i + \lambda \cdot (x_{\text{nn}} - x_i)$$

where $\lambda \in [0, 1]$

Interpolates in feature space
Does not invent new chemistry

Feature Selection: Before vs After SMOTE (Animal Musk)

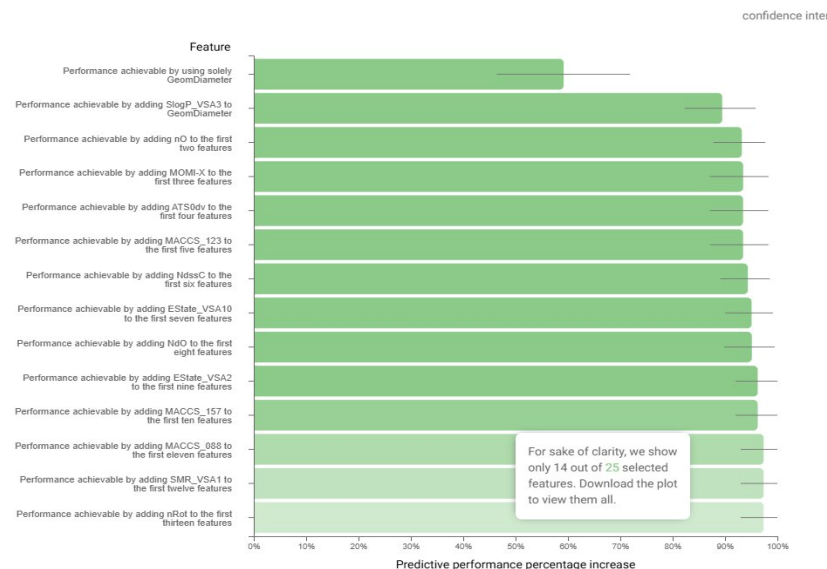
BEFORE SMOTE — Imbalanced



First feature alone (EState_VSA2): 85.7%

Model is exploiting majority class and not learning musk chemistry

AFTER SMOTE — Balanced



First feature (GeomDiameter): 59 % gradual build-up

Model is actually learning Animal Musk chemistry

Why Feature Selection After SMOTE Is Valid

1

Statistical Prior Correction

SMOTE corrects the class probability imbalance in the training distribution. It does not generate new chemical patterns, it rebalances the weight the classifier assigns to each class during training. All chemical information comes entirely from real molecules.

2

Train / Test Separation

SMOTE was applied only to the training split. The test set contains exclusively real molecules. This preserves the integrity of generalization. Jad Bio's evaluation metrics are discarded (computed on SMOTE-augmented data), but this does not affect the validity of feature selection.

3

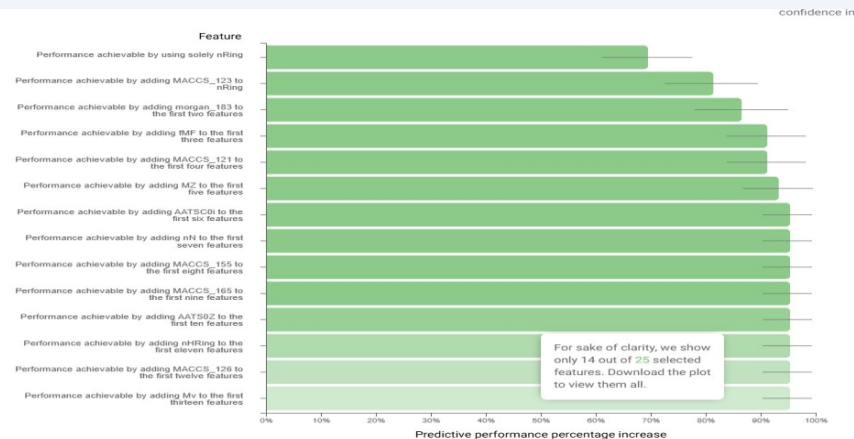
Feature Selection Is Grounded in Real Data

The features selected after balancing are those that genuinely characterize the minority class in the real molecules. Synthetic samples do not introduce new chemical information, they simply ensure the classifier pays equal attention to both classes, preventing the corruption we saw in the imbalanced case.

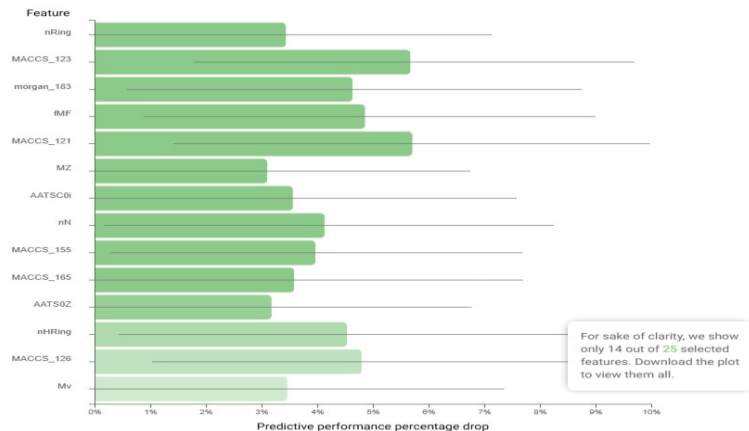
3. Feature Selection Results

Per label analysis : Progressive feature inclusion , feature importance and selected signature

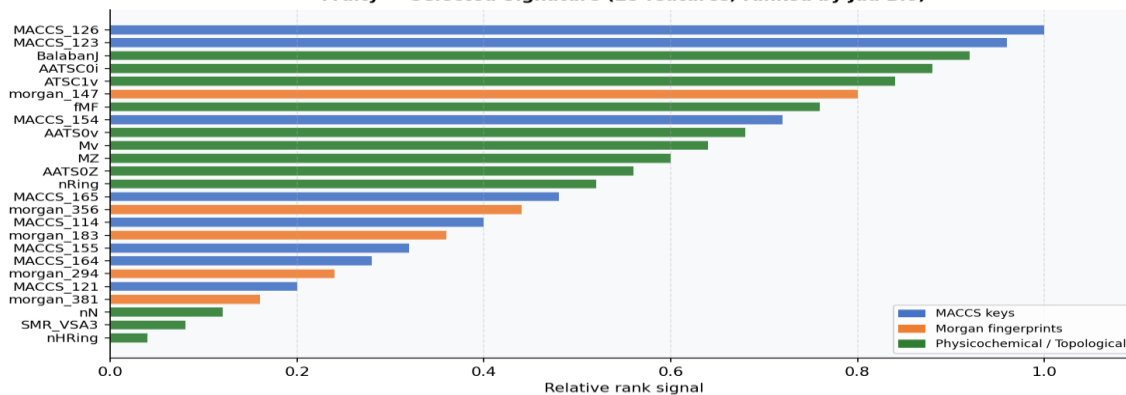
Cumulative Performance



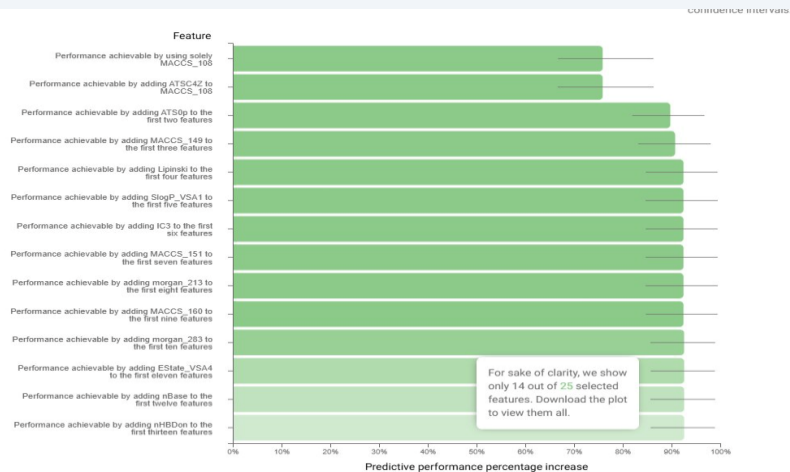
Feature Importance Drop



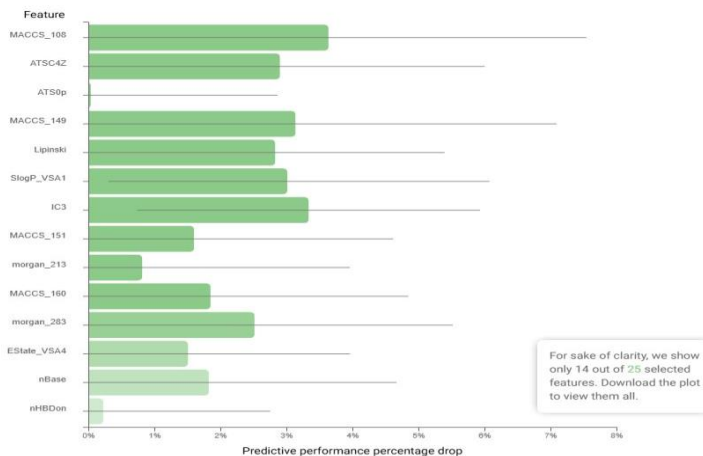
Fruity — Selected Signature (25 features, ranked by Jad Bio)



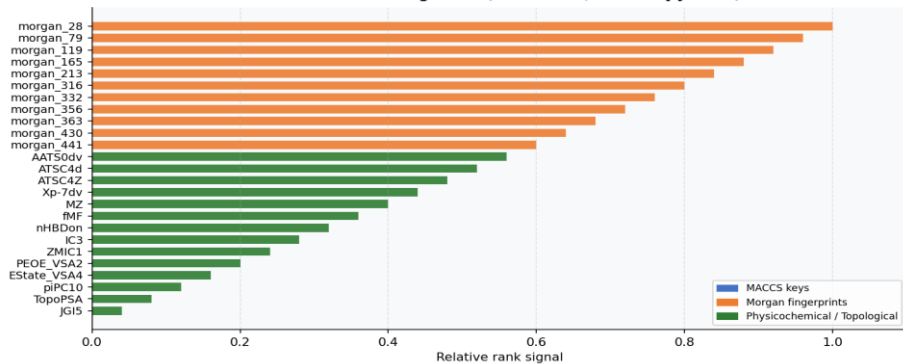
Cumulative Performance



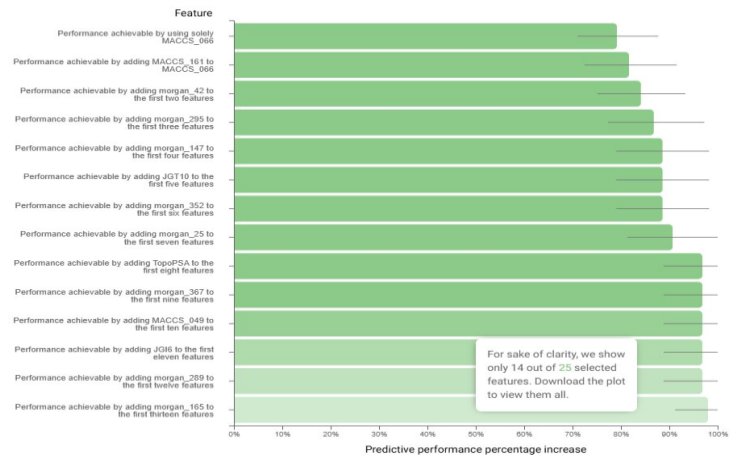
Feature Importance Drop



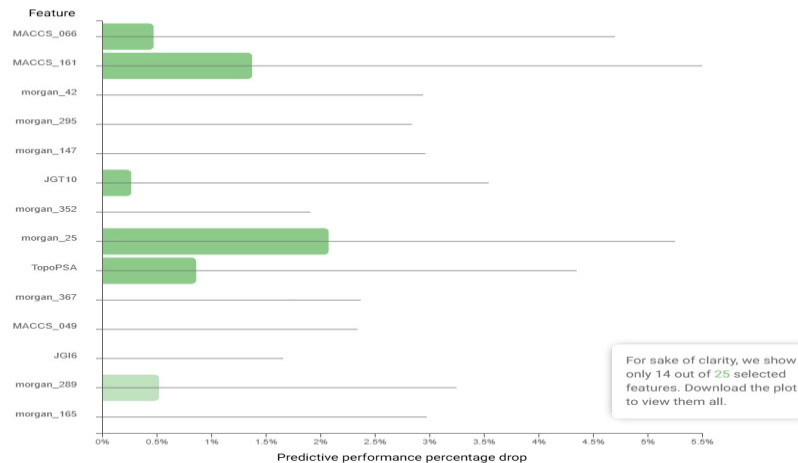
Green — Selected Signature (25 features, ranked by Jad Bio)



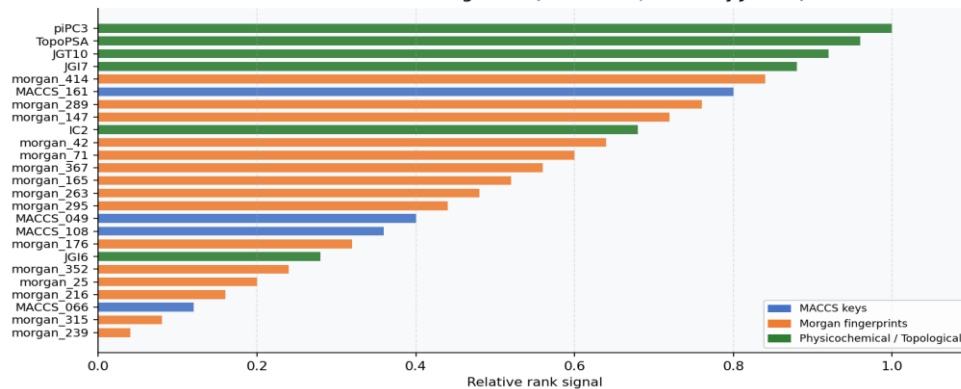
Cumulative Performance



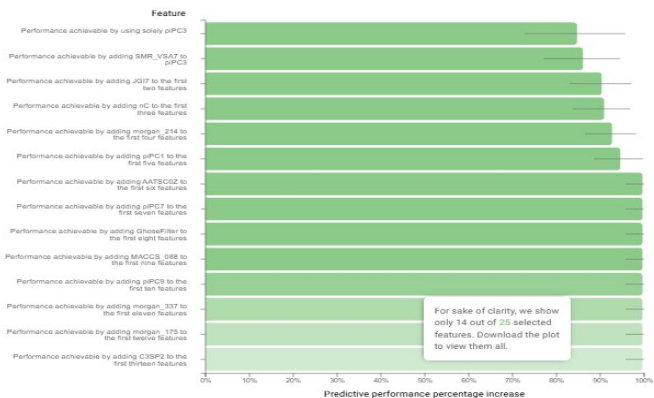
Feature Importance Drop



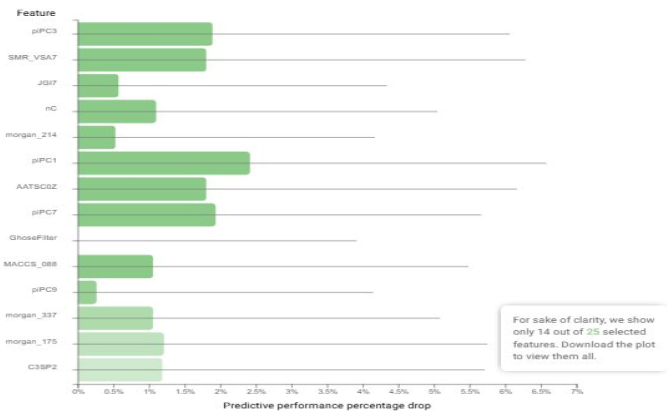
Chemical — Selected Signature (25 features, ranked by Jad Bio)



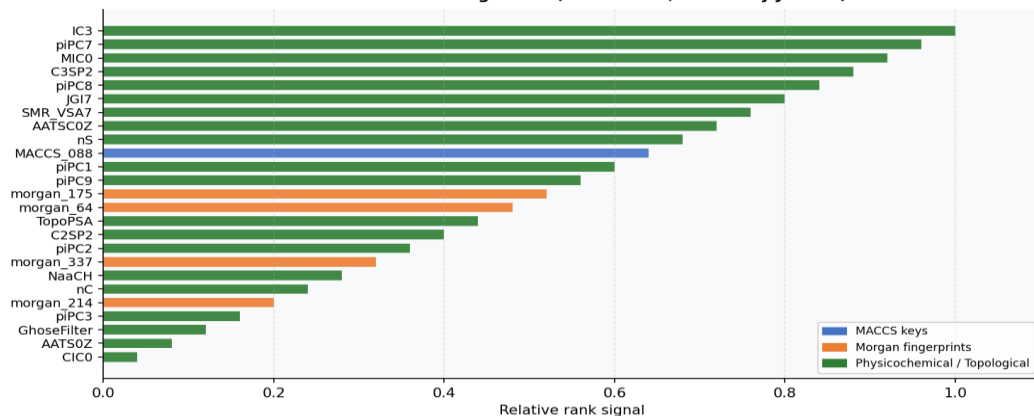
Cumulative Performance



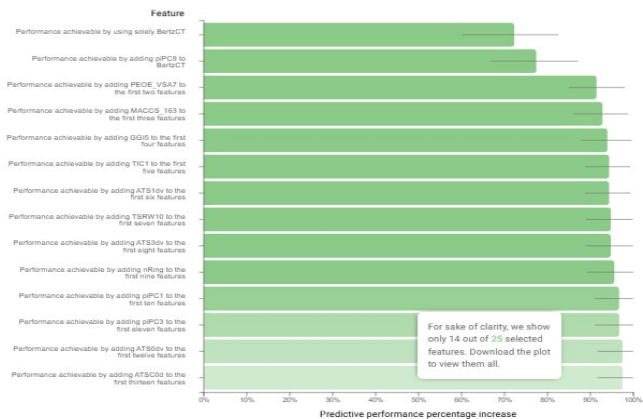
Feature Importance Drop



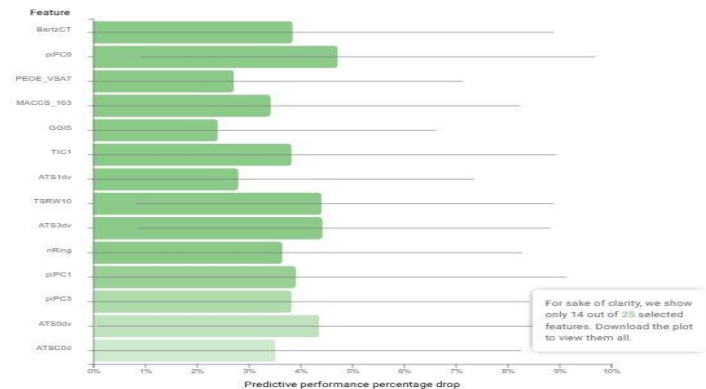
Floral — Selected Signature (25 features, ranked by Jad Bio)



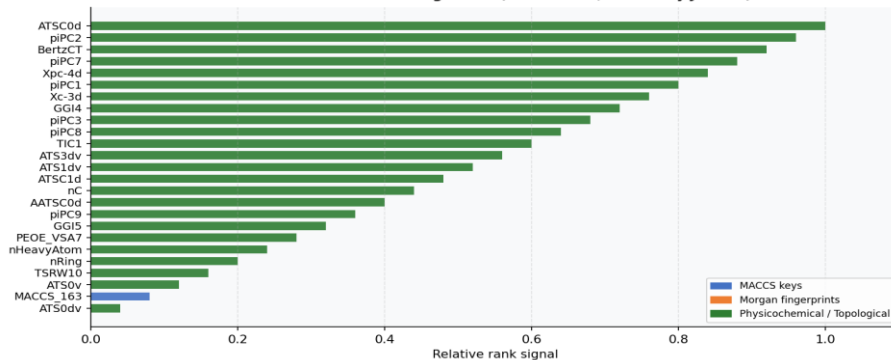
Cumulative Performance



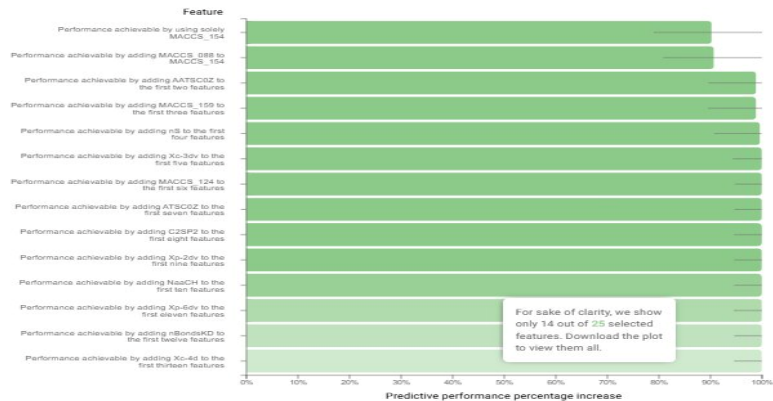
Feature Importance Drop



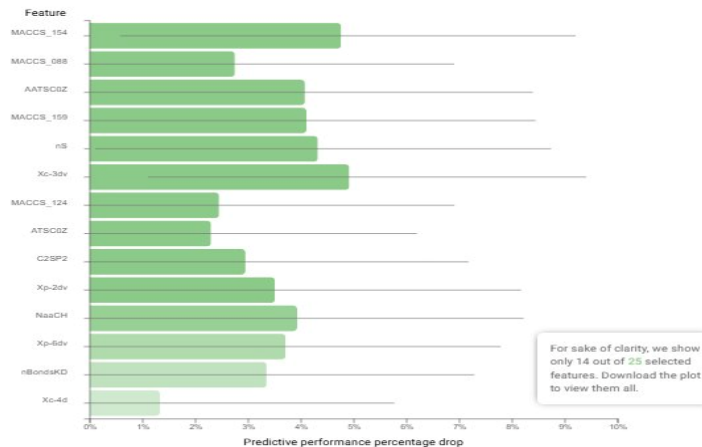
Gourmand — Selected Signature (25 features, ranked by Jad Bio)



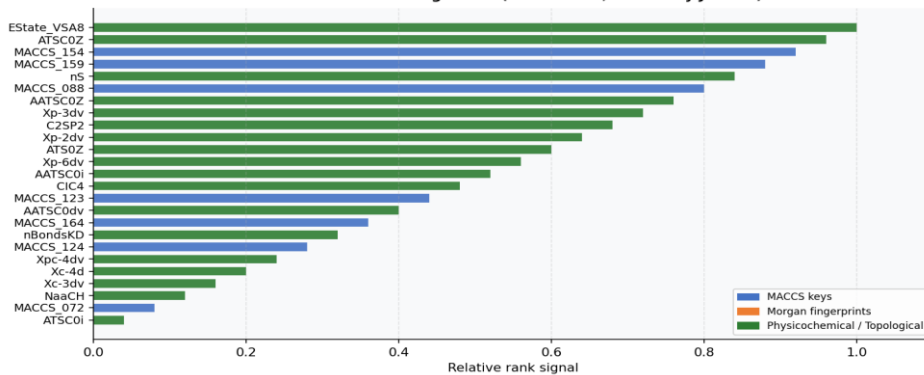
Cumulative Performance



Feature Importance Drop



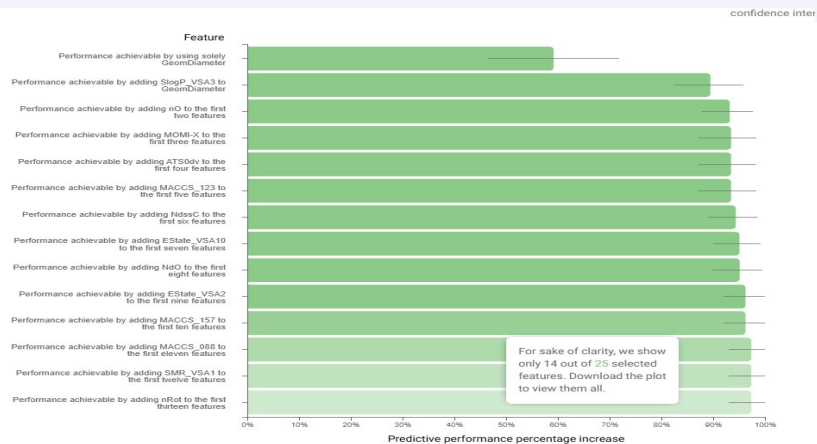
Sweet — Selected Signature (25 features, ranked by Jad Bio)



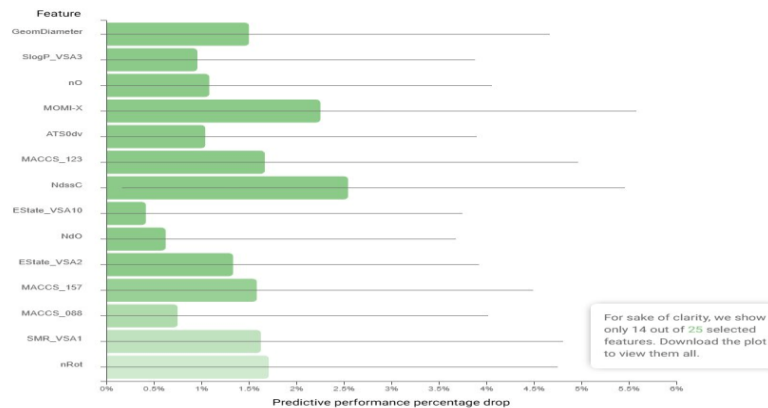
Animal Musk

SMOTE 15.5% → 50%

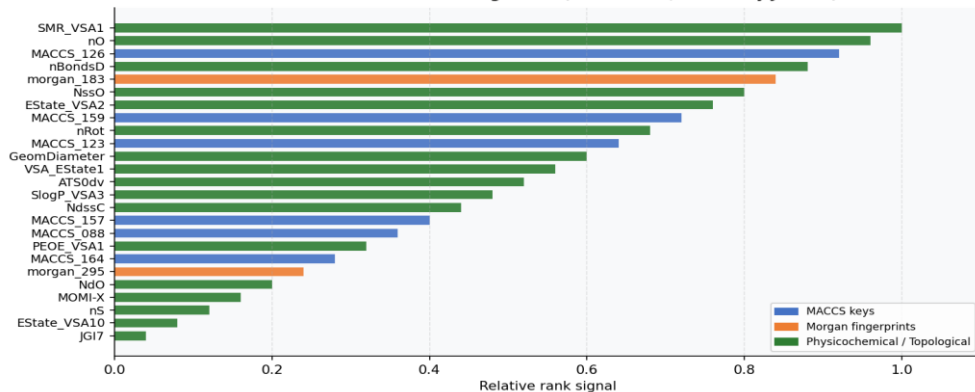
Cumulative Performance



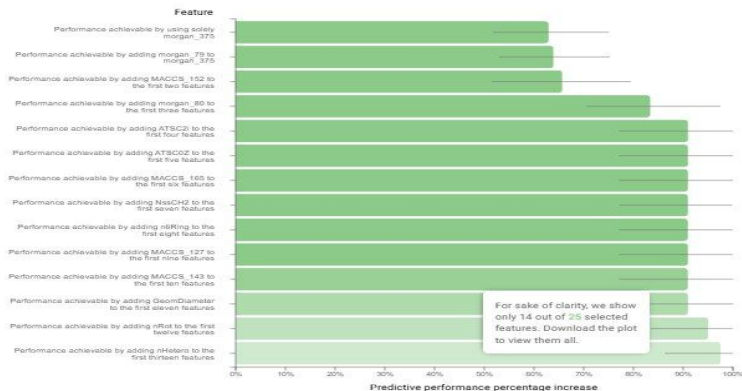
Feature Importance Drop



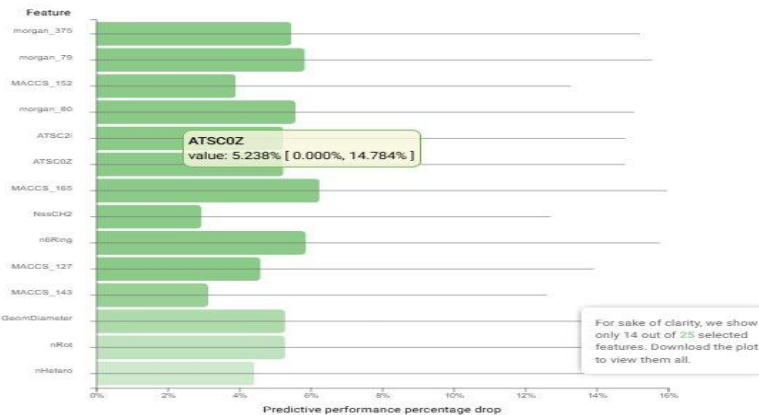
Animal musk — Selected Signature (25 features, ranked by Jad Bio)



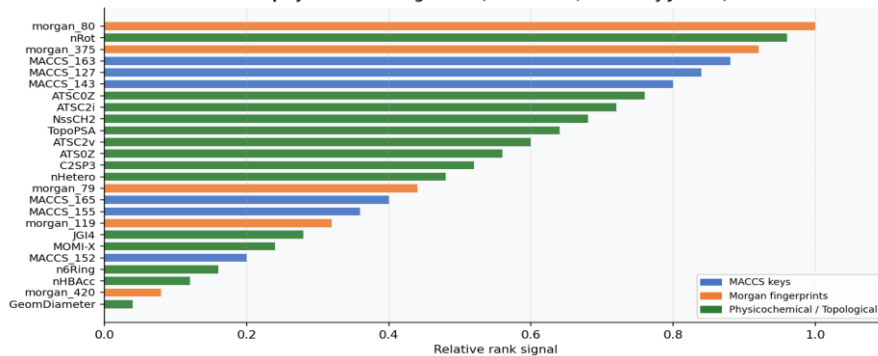
Cumulative Performance



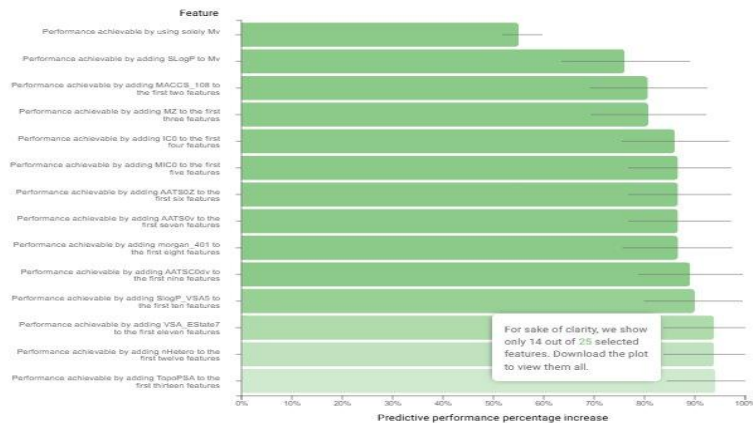
Feature Importance Drop



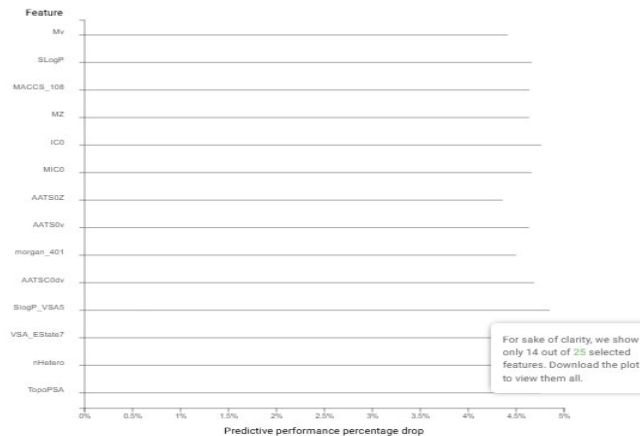
Spicy — Selected Signature (25 features, ranked by Jad Bio)



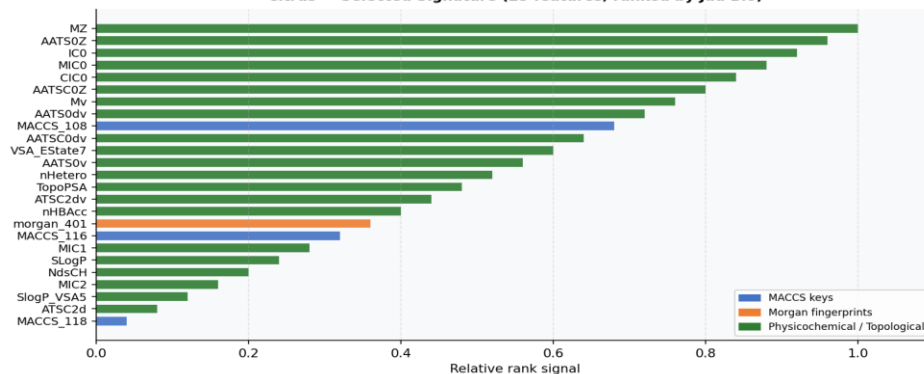
Cumulative Performance



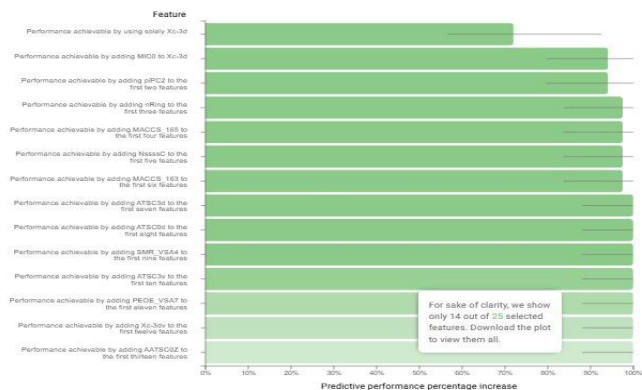
Feature Importance Drop



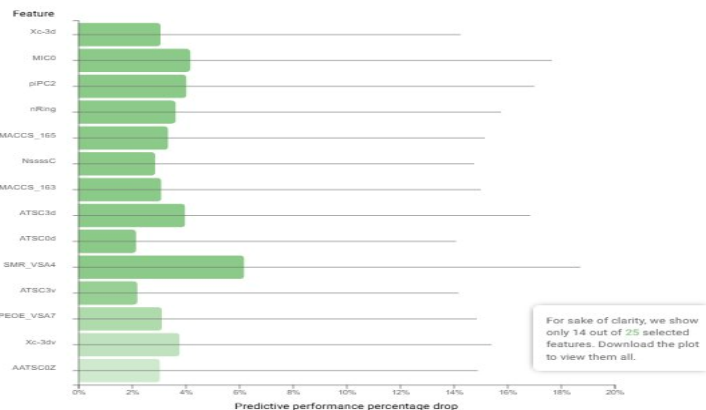
Citrus — Selected Signature (25 features, ranked by Jad Bio)



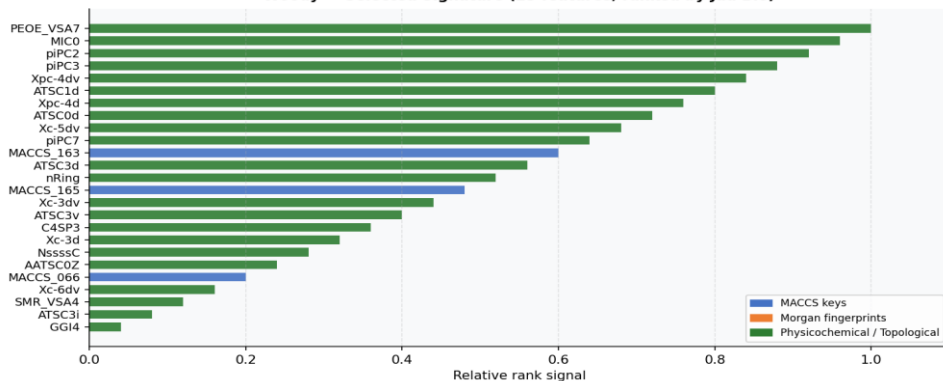
Cumulative Performance



Feature Importance Drop



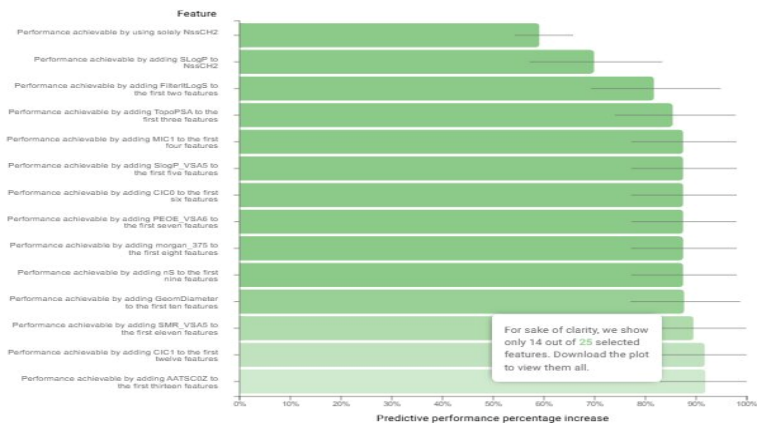
Woody — Selected Signature (25 features, ranked by Jad Bio)



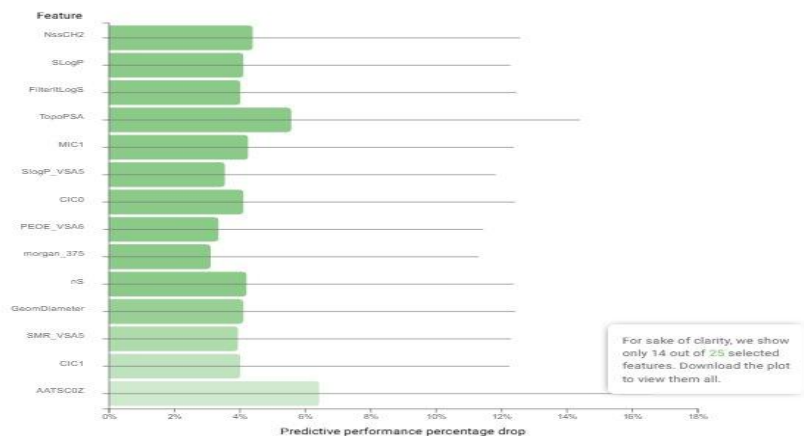
Powdery Amber

SMOTE 20.6% → 50%

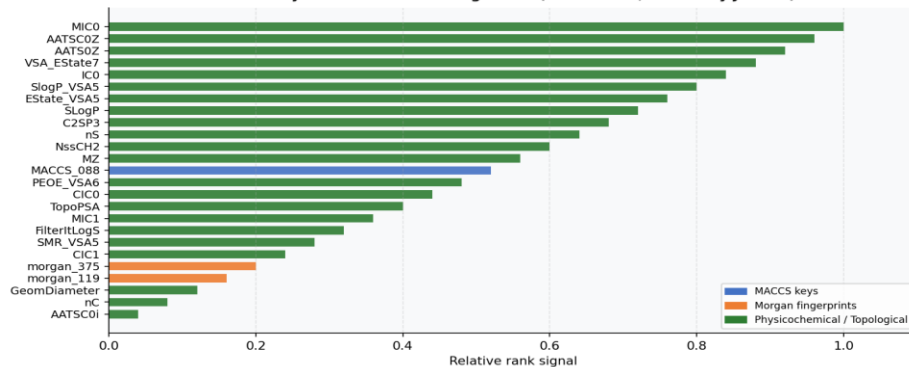
Cumulative Performance



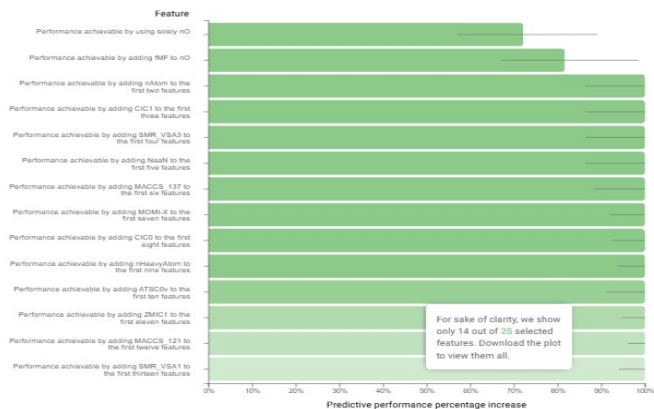
Feature Importance Drop



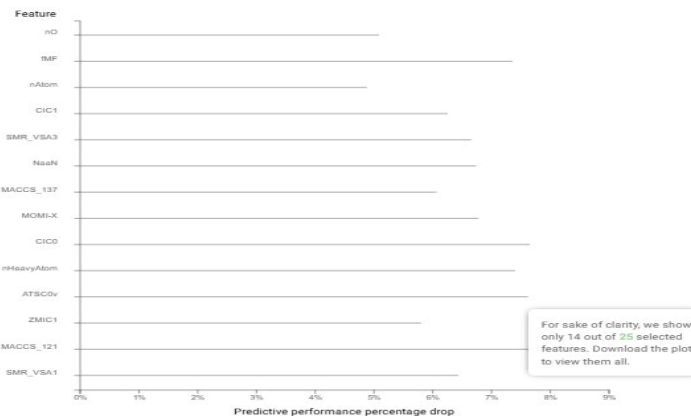
Powdery amber — Selected Signature (25 features, ranked by Jad Bio)



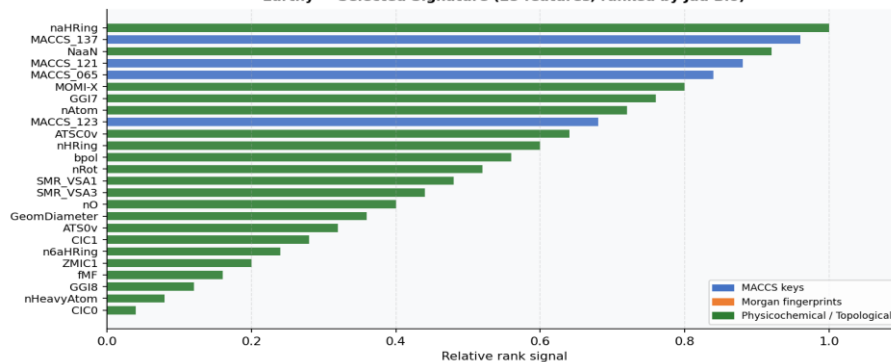
Cumulative Performance



Feature Importance Drop



Earthy — Selected Signature (25 features, ranked by Jad Bio)



4. Cross-Label Feature Analysis

What the 12 models tell us about molecular feature validity

Feature Family Dominance Across Labels		
Label	Dominant Family	What this means
Fruity	MACCS keys	Functional group presence sufficient
Sweet	MACCS keys	C=C alone captures 85% of signal
Green	Morgan fingerprints	Local environments > global groups
Spicy	Morgan + MACCS	family + specific environment
Floral	Topological (piPC)	Unsaturation
Gourmand	Topological (BertzCT/GGI)	Molecular complexity drives prediction
Woody	Topological (Xc/piPC)	Ring topology and branching
Animal Musk	VSA + Geometric	Molecular size, lipophilicity, surface
Powdery Amber	Lipophilicity + Size	Large non-polar molecules
Citrus	Info content (IC/MIC)	Complexity range
Earthy	Ring + Size + MACCS	Aromatic heterocycles + terpene size
Chemical	Mixed	No dominant family

What Class Imbalance Was Hiding

Woody

CRITICAL FINDING

The original conclusion was wrong. 2D features do encode woody structural information.

Before: Pre-SMOTE importance ≈ 0

After: Post-SMOTE: SMR_VSA4, MIC0, piPC2 show 4-6% individual drops

Spicy

VALIDATION

Cross-condition consistency = strongest possible proof that features are genuinely predictive.

Before: Pre-SMOTE: MACCS_143, MACCS_163 were top features

After: Post-SMOTE: MACCS_143, MACCS_163 reappear , same core signature

Floral

VALIDATION

Overlapping signatures across both conditions confirm real chemical signal. Imbalance only suppressed part of the piPC family.

Before: Pre-SMOTE: morgan_337, IC3, C3SP2, TopoPSA, piPC7 selected

After: Post-SMOTE: same features reappear + extended piPC family